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Creating Electronic Books with Code and Artificial Intelligence Assistants

Course for the Faculties of Sciences and Chemical Sciences

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Welcome to **My First Cytology Book**.

This educational resource is designed to serve as an interactive guide for studying and analyzing cytopathologies, providing a dynamic learning experience for health sciences and biology students.

Featured Content

In this interactive book, you will have access to practical, visually enriched content:

- *Image Analysis of Breast Cytologies*: An interactive sample chapter detailing Fine Needle Aspiration Cytology (FNAC) evaluation, correlation of malignancy criteria, and the decision-making algorithm based on the Yokohama System.

Note

This interactive book incorporates dynamic diagrams, comparative tables, and self-assessment tools to enhance the practical learning process.

Part I

Gynecological Cytology

UNIT 5: ANALYSIS OF CERVICOVAGINAL SMEARS IN NEOPLASTIC PROCESSES

(Translation pending / Traducción pendiente)

This chapter covers the evaluation and classification criteria for cervicovaginal intraepithelial lesions and associated carcinomas, focusing on Bethesda nomenclature, low-grade (L-SIL) and high-grade (H-SIL) lesions, squamous cell carcinoma, and glandular abnormalities.

UNIT 7: ANALYSIS OF CYTOLOGICAL SAMPLES OF VULVA, ENDOMETRIUM, TUBES, AND OVARY

(Translation pending / Traducción pendiente)

This chapter covers the cytological and histological evaluation of non-cervical gynecological organs, including vulvar lesions (infections, VIN, squamous cell carcinoma), endometrial cycles, endometritis, endometrial hyperplasia, endometrial adenocarcinoma, fallopian tubes, and ovarian cysts and tumors.

Part II

Breast Cytology

CHAPTER 6: IMAGE ANALYSIS OF BREAST CYTOLOGIES

Fine Needle Aspiration Cytology (FNAC) is a key tool in the diagnosis of breast lesions. Microscopic analysis of the samples allows for the classification of lesions and guides clinical management of the patient rapidly and minimally invasively.

3.1 Cytological Evaluation Criteria

To correctly analyze a breast cytology image, we must focus on three key levels:

1. **Background pattern:** Presence of amorphous material (mucin, necrosis), inflammatory cells, or bipolar naked nuclei (typical of benign lesions like fibroadenomas).
2. **Cellular architecture:** Cohesion of the groups, arrangement in monolayers, or formation of three-dimensional structures (papillae, acini, palisades).
3. **Cytological/nuclear details:** Nucleus-cytoplasm ratio, pleomorphism, nuclear membrane irregularity, and prominence of nucleoli.

3.1.1 Table 1. Correlation of Cytological Criteria in Breast

The main microscopic findings according to the nature of the lesion are summarized below:

Feature	Benign Lesions	Malignant Lesions
Cohesiveness	High (cohesive monolayer sheets)	Low (single cells, loss of cohesion)
Bipolar naked nuclei	Abundant in the background	Absent or very scarce
Nuclear Atypia	Absent or minimal	Moderate to severe (pleomorphism)
Chromatin	Fine, evenly distributed	Coarse, irregular, with nuclear clearing
Nucleoli	Inconspicuous or absent	Prominent, multiple, or giant

3.2 Classification Flow (Yokohama System)

The **Yokohama System** for breast cytopathology provides a standardized guide with five well-defined diagnostic categories.

The following diagram summarizes the Yokohama categories and the recommended clinical approach:

Diagram 1. Yokohama Algorithm for Breast Cytopathology.

As shown in Fig. 3.1, the diagram is versioned as a static image.

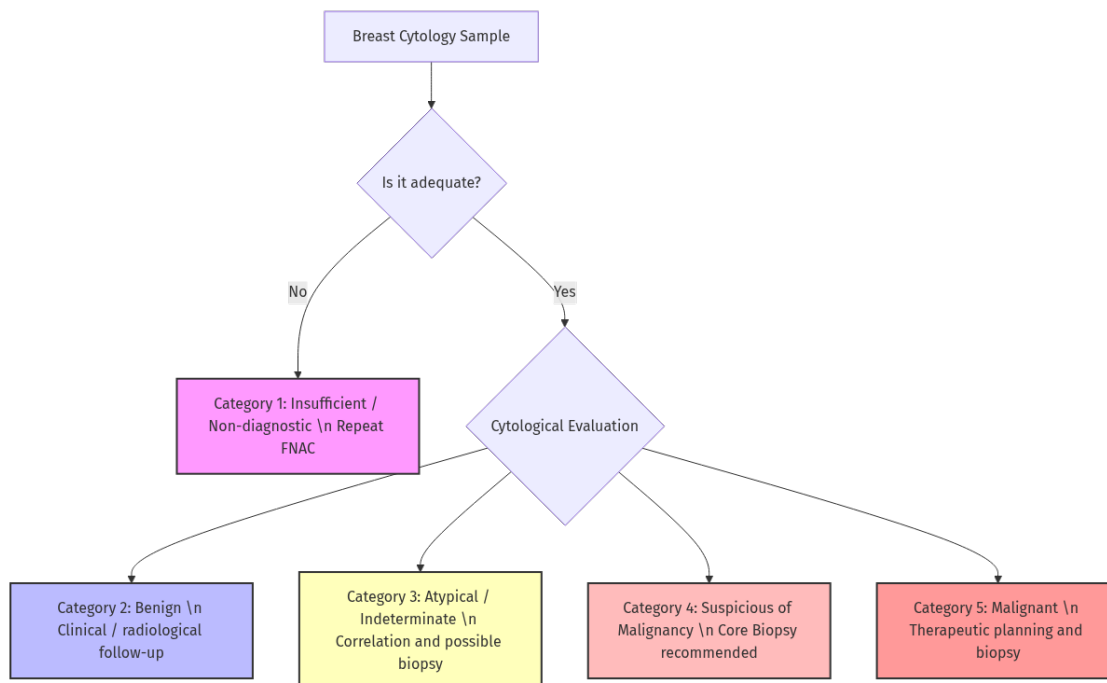


Figure3.1: Diagram 1. Yokohama Algorithm for Breast Cytopathology.

3.3 Interactive Self-Assessment

Now, test yourself with the following self-assessment question based on real cytological cases.

Review Question: Identification of Malignancy Criteria

In a breast FNAC cytology, we observe abundant single (dissociated) epithelial cells with marked nuclear pleomorphism, irregular nuclear membranes, and prominent nucleoli. Bipolar naked nuclei are not observed in the background.

What is the most likely cytological diagnosis and which category of the Yokohama System does it correspond to?

i View Solution and Discussion

Correct Answer: Malignant Lesion (Category 5 of the Yokohama System).

Discussion:

1. **Loss of cohesiveness:** The presence of abundant single (dissociated) cells is a classic criterion of malignancy (for example, in infiltrating ductal carcinoma). Benign lesions (such as fibroadenoma) typically show highly cohesive cell sheets with a “staghorn” appearance.
2. **Marked nuclear atypia:** Pleomorphism, nuclear membrane irregularity, and prominent nucleoli indicate a profound alteration of cellular homeostasis.
3. **Absence of bipolar naked nuclei:** Benign bipolar nuclei represent healthy myoepithelial cells shedding from the stroma. Their absence in a cellular sample strongly supports a malignant diagnosis.

UNIT 6: DETAILED IMAGE ANALYSIS OF BREAST CYTOLOGY

(Translation pending / Traducción pendiente)

This chapter covers the anatomical and histological features of the breast, cytology examination methods (FNAC, imprint, nipple discharge, core needle biopsy), normal cytology patterns (epithelial, myoepithelial, stromal bipolar naked nuclei, apocrine metaplasia), non-tumoral pathology (mastitis, fat necrosis), and benign and malignant breast tumors (fibroadenoma, ductal carcinoma, lobular carcinoma).

Part III

Image Atlas

DIAGNOSTIC IMAGE ATLAS

(Translation pending / Traducción pendiente)

This atlas gathers key microscopic images for gynecological and breast cytopathology studies, organized by teaching units (Unit 1, Unit 3, Unit 4, Unit 5).

Part IV

Information

ABOUT

This book contains the course material for **Elaboración de libros electrónicos mediante código y asistentes de Inteligencia Artificial**. It is designed to help teaching staff at the Faculties of Sciences and Chemical Sciences of the University of Salamanca create, adapt and publish interactive teaching books with minimal technical friction.

Citation information, BibTeX, DOI and Zenodo metadata are collected on the *How to Cite* page.

6.1 How this book is made

This website is written in Markdown files and Jupyter notebooks, and converted to HTML and PDF using tools from [TeachBooks](#), [Jupyter Book](#) and [MyST Markdown](#).

The source files are stored in a public GitHub repository:

- https://github.com/elloza/teachbook_usal_template

The published version of the book can be viewed at:

- https://elloza.com/teachbook_usal_template/

6.2 About the authors



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6.2.1 Institution

Faculties of Sciences and Chemical Sciences, University of Salamanca (USAL).

Photographs and academic profile information come from the University of Salamanca scientific production portal.

CREDITS AND LICENSES

7.1 Project credits

This book and the course template are built with [TeachBooks](#), on top of the [Jupyter Book](#) and [MyST Markdown](#) ecosystem.

TeachBooks provides documentation, components and working patterns that make it easier to create open, interactive and reproducible teaching books.

7.2 Acknowledgements

We thank the [TeachBooks](#) project and [Delft University of Technology \(TU Delft\)](#) for their work on open tools for educational books. The [credits page in the TeachBooks Manual](#) is used as a reference for explicitly acknowledging the tools, communities and institutional support on which this material builds.

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7.5 Code and tools

This project uses external tools and libraries, including:

- [TeachBooks](#)
- [Jupyter Book](#)
- [MyST Markdown](#)

Each tool keeps its own license.

HOW TO CITE

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If you use this material, please cite it as follows.

8.1 Recommended Citation (APA 7th edition)

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8.2 BibTeX

You can also use the downloadable BibTeX file: `citation.bib`.

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